Ch04: Prediction, Goodness-of-fit, and Modeling Issues: 4.1-4.2

Financial Econometrics

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4.1 Least Square Prediction

Fig 4.1 Point prediction

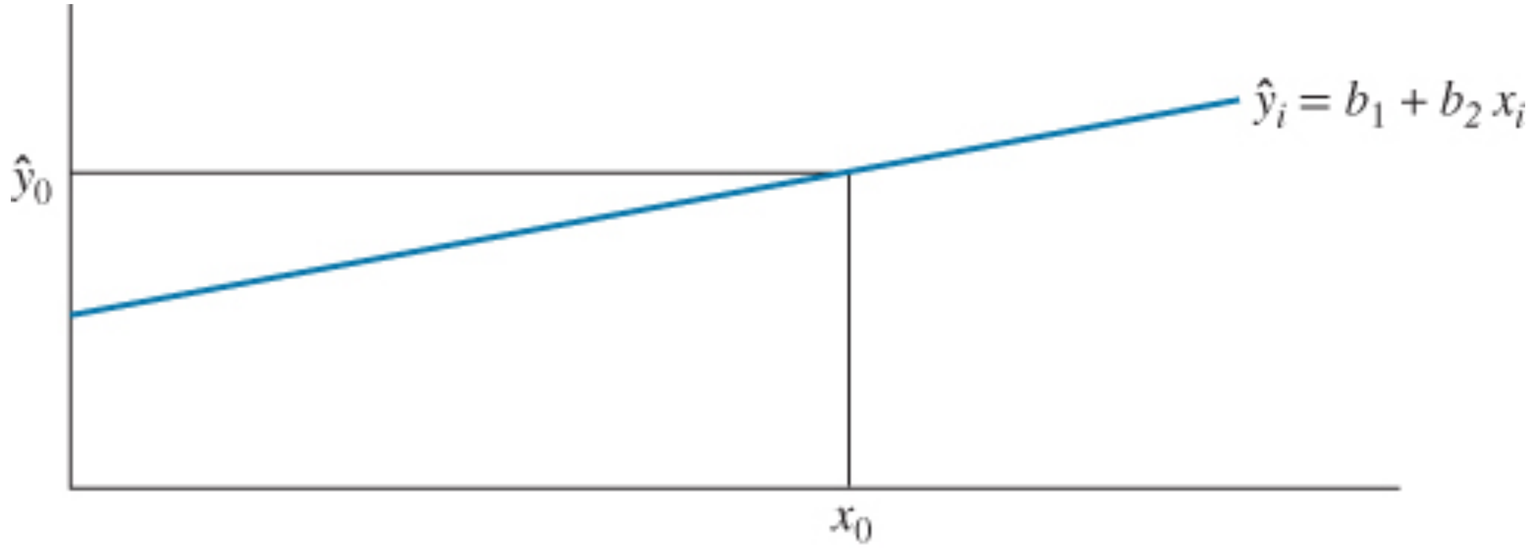
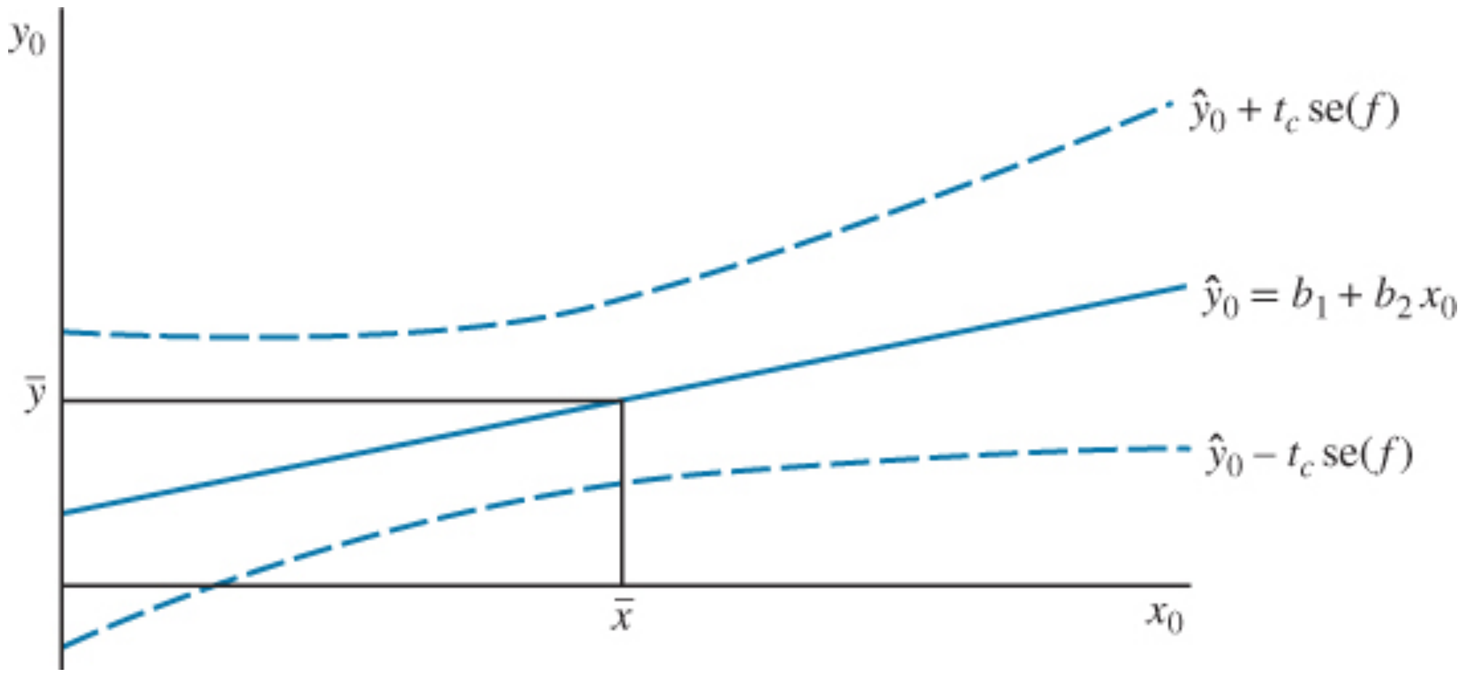


Fig 4.2. Point and interval prediction



The least squares predictor

To do prediction, assume y_0 and x_0 are related to one another by the same regression model,

$$y_0 = \beta_1 + \beta_2 x_0 + e_0,$$

where e_0 is a random error.

The least squares predictor of y_0 comes from the fitted regression line

$$\hat{y}_0 = b_1 + b_2 x_0.$$

The forecast error

To evaluate how well this predictor performs, we define the forecast error by

$$f = y_0 - \hat{y}_0 = (\beta_1 + \beta_2 x_0 + e_0) - (b_1 + b_2 x_0)\,.$$

Because f is a linear combination of y_i and an additional random error e_0),
$f \sim N(E(f) = 0, var(f))$ with

$$var(f) = \sigma^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}\,. \right] \tag{1}$$

$var(f)$ is proved in the appendix.

Because $E(f) = 0$, $\hat{y}_0$ is an unbiased predictor of y_0.

By the variance of this predictor in (1), the variance of the forecast is smaller when

1. the overall uncertainty σ^2 is smaller;
2. the sample N is larger;
3. the variation in the explanatory variable is larger;
4. the value of $(x_0 - \bar{x})^2$ is small.

$\hat{y}_0$ is the best linear unbiased predictor of y_0. [Proof is skipped.]

To estimate the variance of forecast error

For estimating variance, we use

$$\hat{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right].$$

Let the standard error of the forecast be $se(f) = \sqrt{\hat{var}(f)}$.

Because of $f = (y_0 - \hat{y}) \sim N(0, var(f))$, we have

$$\frac{f - 0}{se(f)} = \frac{y_0 - \hat{y}_0}{se(f)} \sim t_{N-2}.$$

Prediction interval

Similar to the confidence interval,

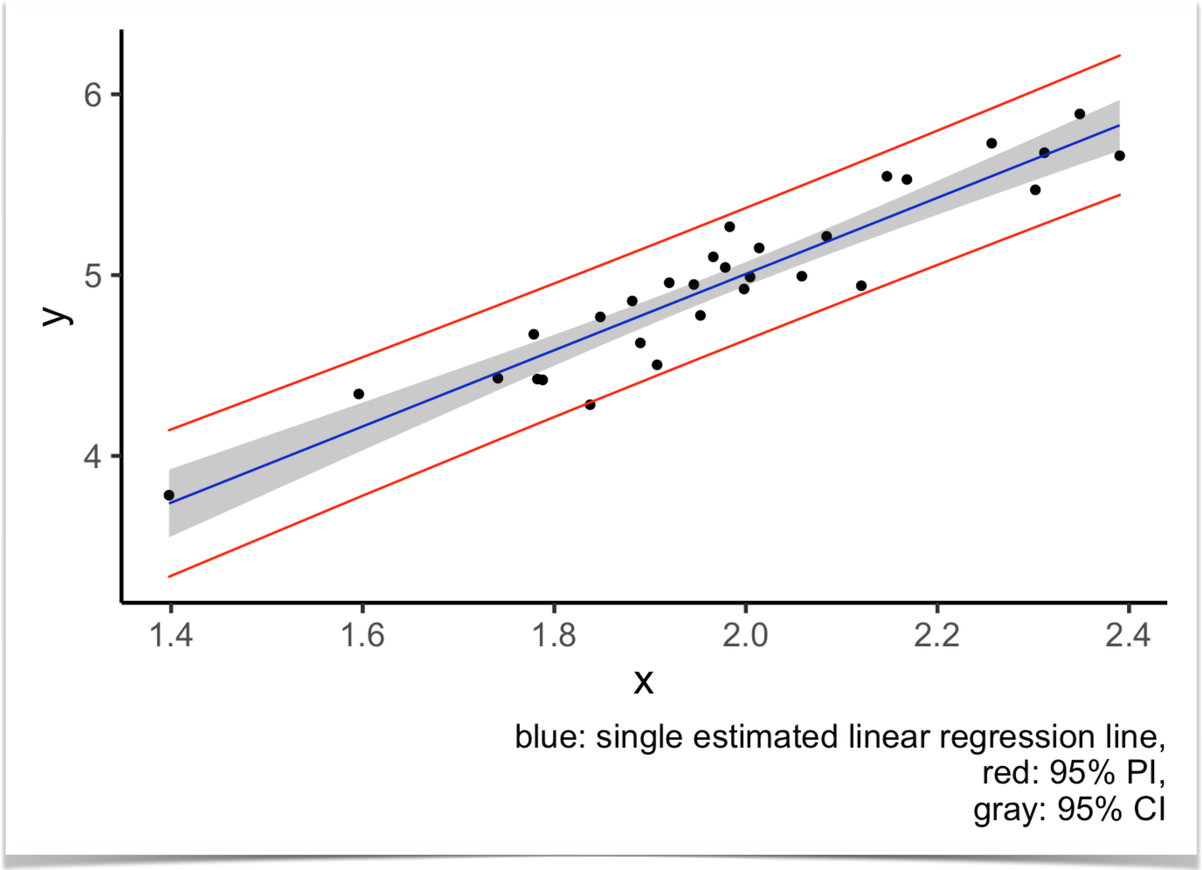
$$
\begin{aligned}
1-\alpha &= P\left(-t_c \le \frac{y_0-\hat{y}_0}{se(f)} \le t_c\right) \\
&= P\left(-t_c se(f) \le y_0-\hat{y}_0 \le t_c se(f)\right) \\
&= P\left(\hat{y}_0 - t_c se(f) \le y_0 \le \hat{y}_0 + t_c se(f)\right)
\end{aligned}
$$

where $t_c = t_{(N-2),\alpha/2}$, we define the $100(1-\alpha)$ *prediction interval* for y_0 as

$$
[\hat{y}_0 - t_c se(f), \hat{y}_0 + t_c se(f)] = [(b_1+b_2x_0) - t_c\hat{\sigma}\sqrt{[1+\frac{1}{N}+\frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}]}, \quad (b_1+b_2x_0) + t_c\hat{\sigma}\sqrt{[1+\frac{1}{N}+\frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}]}].
$$

In R, we use qt(1-alpha/2, N-2) to calculate t_c.

Difference between a CI for $E[y \mid x]$ and a PI for y given $x = x_0$



https://www.kevinwangstats.com/post/2021-05-30-confidence-and-prediction-intervals/

Comparisons between the PI and CI for $\beta_1 + \beta_2 x$

Special case: CI for $\beta_1 + \beta_2 x$

where $t_c = t_{(N-2),\alpha/2}$, we define the $100(1-\alpha)$ *prediction interval* for y_0 as

$$[\hat{y}_0 - t_c se(f), \hat{y}_0 + t_c se(f)] = [(b_1 + b_2 x) - t_c \hat{\sigma}\sqrt{[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}]}, \quad (b_1 + b_2 x) + t_c \hat{\sigma}\sqrt{[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}]}].$$

$$[\hat{y} - t_c se(\hat{y}), \quad \hat{y} + t_c se(\hat{y})]$$

$$[(b_1 + b_2 x) - t_c \hat{\sigma}\sqrt{[\frac{1}{N} + \frac{(x - \bar{x})^2}{\sum (x_i - \bar{x})^2}]}, \quad (b_1 + b_2 x) + t_c \hat{\sigma}\sqrt{[\frac{1}{N} + \frac{(x - \bar{x})^2}{\sum (x_i - \bar{x})^2}]}]$$

$e(\hat{y})$

R

Calculate from scratch

```
data("food")
alpha <- 0.05
x <- 20
xbar <- mean(food$income)
m1 <- lm(food_exp~income, data=food)
b1 <- coef(m1)[[1]]
b2 <- coef(m1)[[2]]
yhatx <- b1+b2*x
sm1 <- summary(m1)
df <- df.residual(m1)
tcr <- qt(1-alpha/2, df)
N <- nobs(m1)   #number of observations, N
N <- NROW(food) #just another way of finding N
varb2 <- vcov(m1)[2, 2]
sighat2 <- sm1$sigma^2 # estimated variance
```

R

```
varf <- sighat2+sighat2/N+(x-xbar)^2*varb2 #forecast variance
sef <- sqrt(varf) #standard error of forecast
lb <- yhatx-tcr*sef
ub <- yhatx+tcr*sef
```

The prediction interval for the forecast is (104.13, 471.09).

A different way of finding point and interval estimates for the predicted $E(y \mid x)$ and forecasted y in R:

```
incomex=data.frame(income=20)
predict(m1, newdata=incomex, interval="confidence",level=0.95)
predict(m1, newdata=incomex, interval="prediction",level=0.95)
```

Use predict(.)
1) Confidence Interval
2) Prediction Interval

Appendix: Proof of $var(f)$

Skipped in class. Please read over it and talk with me if you have questions.

Appendix: Proof of $var(f)$

Recall that the forecast error is:

$$f = y_0 - \hat{y}_0 = (\beta_1 + \beta_2 x_0 + e_0) - (b_1 + b_2 x_0) \sim N(E(f), var(f)) \, .$$

- First, it is easy to see $E(f) = 0$.

- Second, we have

$$
\begin{aligned}
\mathrm{var}(\hat{y}_0) &= \mathrm{var}(b_1 + b_2 x_0) \\
&= \mathrm{var}(b_1) + x_0^2 \mathrm{var}(b_2) + 2x_0 \mathrm{cov}(b_1, b_2) \\
&= \frac{\sigma^2 \sum x_i^2}{N \sum (x_i - \bar{x})^2} + x_0^2 \frac{\sigma^2}{\sum (x_i - \bar{x})^2} + 2x_0 \sigma^2 \frac{-\bar{x}}{\sum (x_i - \bar{x})^2} \\
&= \left[\frac{\sigma^2 \sum x_i^2}{N \sum (x_i - \bar{x})^2} - \left\{ \frac{\sigma^2 N \bar{x}^2}{N \sum (x_i - \bar{x})^2} \right\} \right] \\
&+ \left[\frac{x_0^2 \sigma^2}{\sum (x_i - \bar{x})^2} + \frac{\sigma^2 (-2x_0 \bar{x})}{\sum (x_i - \bar{x})^2} + \left\{ \frac{\sigma^2 N \bar{x}^2}{N \sum (x_i - \bar{x})^2} \right\} \right] \\
&= \sigma^2 \left[\frac{\sum x_i^2 - N\bar{x}^2}{N \sum (x_i - \bar{x})^2} + \frac{x_0^2 - 2x_0 \bar{x} + \bar{x}^2}{\sum (x_i - \bar{x})^2} \right] \\
&= \sigma^2 \left[\frac{\sum (x_i - \bar{x})^2}{N \sum (x_i - \bar{x})^2} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = \sigma^2 \left[\frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]
\end{aligned}
$$

- Therefore, we have

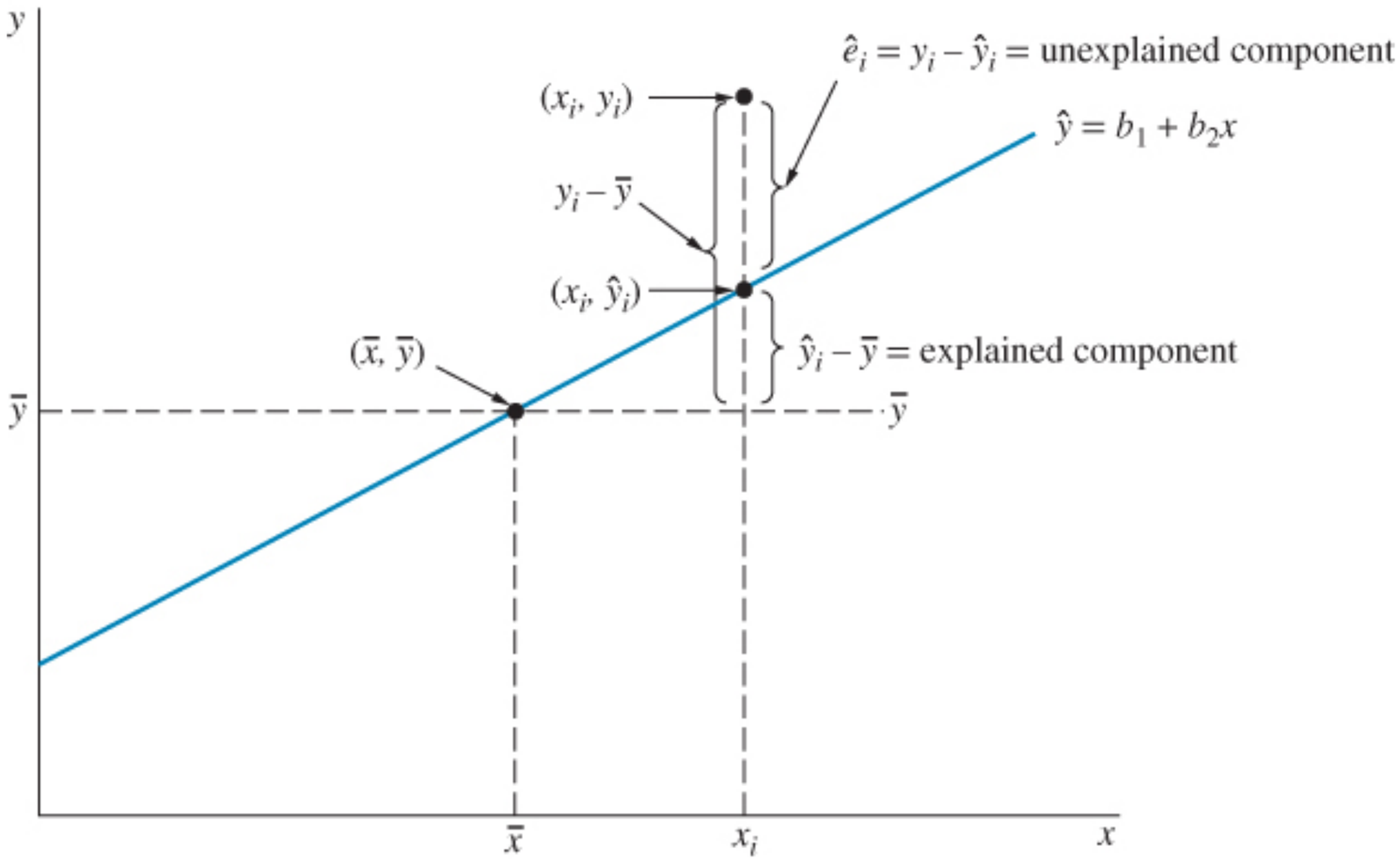
$$var(f) = var(y_0 - \hat{y}_0) = var(e_0 - \hat{y}_0) = var(\hat{y}_0) + \sigma^2,$$

or,

$$var(f) = \sigma^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} . \right]$$

4.2 Measuring Goodness-of-fit

Fig 4.3 Explained and unexplained components of y_i



SST, SSR, SSE

Recall that the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, and the fitted value $\hat{y}_i = b_1 + b_2 x_i$, and the residual $\hat{e}_i = y_i - \hat{y}_i$.

SST = total sum of squares = $\sum (y_i - \bar{y})^2$

SSR = sum of squares due to regression = $\sum (\hat{y}_i - \bar{y})^2$

SSE = sum of squares due to error = $\sum \hat{e}_i^2$

We have

$$\text{SST} = \text{SSR} + \text{SSE}\,.$$

Why?

We write $y_i = \hat{y}_i + \hat{e}_i$, and subtract $\bar{y}$,

$$(y_i - \bar{y}) = (\hat{y}_i - \bar{y}) + \hat{e}_i \,.$$

Squaring and summing over i, with the fact $\sum (\hat{y}_i - \bar{y})\hat{e}_i = 0$ (proved in the appendix), we have

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{e}_i^2 + 2\sum (\hat{y}_i - \bar{y})\hat{e}_i = \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{e}_i^2 \,.$$

R^2

Define the *coefficient of determination*, or, R^2, as the proportion of variation in y explained by x within the regression model:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{SSE}{SST}.$$

When $0 < R^2 < 1$, R^2 is interpreted as "the proportion of the variation in y about its mean that is explained by the regression model".

R demo

Calculate R^2 in the food expenditure example

```
(rsq <- sm1$r.squared) #or sm1
```

If you need SSR or SSE, use ANOVA

```
anov <- anova(m1)
dfr <- data.frame(anov)
kable(dfr,
  caption="Output generated by the ‘anova‘ function")
SSE = anov[2,2]; SSR = anov[1,2]; SST = SSE+SSR;
```

The key ingredients in a report

1. the coefficient estimates
2. the standard errors (or *t*-values)
3. an indication of statistical significance
4. R^2

```
Call:
lm(formula = food_exp ~ income, data = food)

Residuals:
     Min       1Q   Median       3Q      Max 
-223.025  -50.816   -6.324   67.879  212.044 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   83.416     43.410   1.922   0.0622 .  
income        10.210      2.093   4.877 1.95e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 89.52 on 38 degrees of freedom
Multiple R-squared:  0.385,	Adjusted R-squared:  0.3688 
F-statistic: 23.79 on 1 and 38 DF,  p-value: 1.946e-05
```

r_{xy}

The correlation coefficient ρ_{xy} between x and y is defined as

$$\rho_{xy} = \frac{cov(x, y)}{\sqrt{var(x)}\sqrt{var(y)}} .$$

Substituting sample values, we get the sample correlation coefficient:

$$r_{xy} = \frac{s_{xy}}{s_x s_y},$$

where

$$\begin{aligned} s_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y})/(N-1), \\ s_x &= \sqrt{\sum (x_i - \bar{x})^2/(N-1)}, \\ s_y &= \sqrt{\sum (y_i - \bar{y})^2/(N-1)} . \end{aligned}$$

The sample correlation coefficient r_{xy} has a value between -1 and 1, and it measures *the strength of linear association* between observed values of x and y.

$$R^2 = r_{xy}^2$$

- Note that r_{xy}^2 measures the strength of the linear association between x and y.
- This interpretation is not far from that of R^2: the proportion of variation in y about its mean explained by x in the linear regression model.

Prof (from Stat 101)

How to calculate ANOVA table?

1. Total SS $= \sum (y - \bar{y})^2 = S_{yy}$.
2. SSR

$$\text{SSR} := \sum (\hat{y}_i - \bar{y})^2 = \sum ((\boxed{a} + bx_i) - \bar{y})^2 = \sum ((\boxed{\bar{y} - b\bar{x}}) + bx_i - \bar{y})^2$$

$$= \sum (b(x_i - \bar{x}))^2 = b^2 S_{xx} = \left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}.$$

3. SSE = Total SS − SSR

Why $R^2 = r^2$?

- In fact, $R^2 = r^2$, because

$$r^2 = \left(\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}\right)^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \frac{S_{xy}^2}{S_{xx}} \frac{1}{S_{yy}} = \frac{\text{SSR}}{\text{Total SS}} = R^2$$

-

Recall that SSR $= \frac{S_{xy}^2}{S_{xx}}$!

$$R^2 = \frac{SSR}{SST}$$

$$R^2 = r_{y\hat{y}}^2$$

- R^2 can be computed as the square of the sample correlation coefficient between y_i and $\hat{y}_i = b_1 + b_2 x_i$.
- R^2 measures the linear association, or goodness-of-fit, between the sample data and their predicted values.
- Consequently, R^2 is sometimes called a measure of "goodness-of-fit".

Why? This is because $r_{y\hat{y}} = \frac{b}{|b|} r_{xy}$

$$r_{y\hat{y}} = \frac{S_{y\hat{y}}}{\sqrt{S_{yy} S_{\hat{y}\hat{y}}}} = \frac{b S_{yx}}{\sqrt{b^2 S_{yy} S_{xx}}} = \frac{b}{|b|} r_{xy}$$

$$\begin{aligned} S_{y\hat{y}} &= \sum (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}}) \\ &= \sum (y_i - \bar{y})((a + bx_i) - (a + b\bar{x})) \\ &= b \sum (y_i - \bar{y})(x_i - \bar{x}) \end{aligned}$$

$$\begin{aligned} S_{\hat{y}\hat{y}} &= \sum (\hat{y}_i - \bar{\hat{y}}_i)(\hat{y}_i - \bar{\hat{y}}_i) \\ &= \sum ((a + bx_i) - (a + b\bar{x}))((a + bx_i) - (a + b\bar{x})) \\ &= \sum b^2 (x_i - \bar{x})(x_i - \bar{x}) \\ &= b^2 S_{xx} \end{aligned}$$

General R^2

It is a general rule that the squared sample correlation between y and its fitted value $\hat{y}$ is a valid measure of good-of-fit.

A general goodness-of-fit measure, or general R^2, is defined as

$$R_g^2 = [corr(y, \hat{y})]^2 = r_{y\hat{y}}^2 \,.$$

Appendix: Proof of $\sum (\hat{y}_i - \bar{y})\hat{e}_i = 0$

Show that $\sum(\hat{y}_i - \bar{y})\hat{e}_i = 0$ under the linear regression model: $y_i = \beta_1 + \beta_2 x_i + e_i$, where $e_i \sim^{i.i.d.} N(0,\sigma^2)$.
To find the least squared estimates for β_1 and β_2, denoted by b_1 and b_2, respectively, we define

$$S(\beta_1, \beta_2) = \sum (y_i - \beta_1 - \beta_2 x_i)^2,$$

and b_1 and b_2 are solutions for the following first-order-conditions,

$$\sum (y_i - \beta_1 - \beta_2 x_i) = 0,$$
$$\sum x_i(y_i - \beta_1 - \beta_2 x_i) = 0.$$

Plugging b_1 and b_2 into the above equations, we obtain

$$\sum (y_i - b_1 - b_2 x_i) = 0,$$
$$\sum x_i(y_i - b_1 - b_2 x_i) = 0.$$

Using the definition of residuals $\hat{e}_i = y_i - b_1 - b_2 x_i$, we have equal expressions,

$$\sum \hat{e}_i = 0,$$

$$\sum x_i e_i = 0.$$

Hence,

$$\sum (\hat{y}_i - \bar{y})\hat{e}_i = \sum (b_1 + b_2 x_i)\hat{e}_i + \bar{y} \sum \hat{e}_i = b_1 \sum \hat{e}_i + b_2 \sum x_i \hat{e}_i + 0 = 0.$$

4.3 Modelling issues

Transforming the variables

1. Change the scale of x or y.
2. Power: quadratic (t^2), cubic (t^3)
3. Nature logarithm: $ln(t)$.

Fig 4.4 A non-constant relationship between food expenditure and income

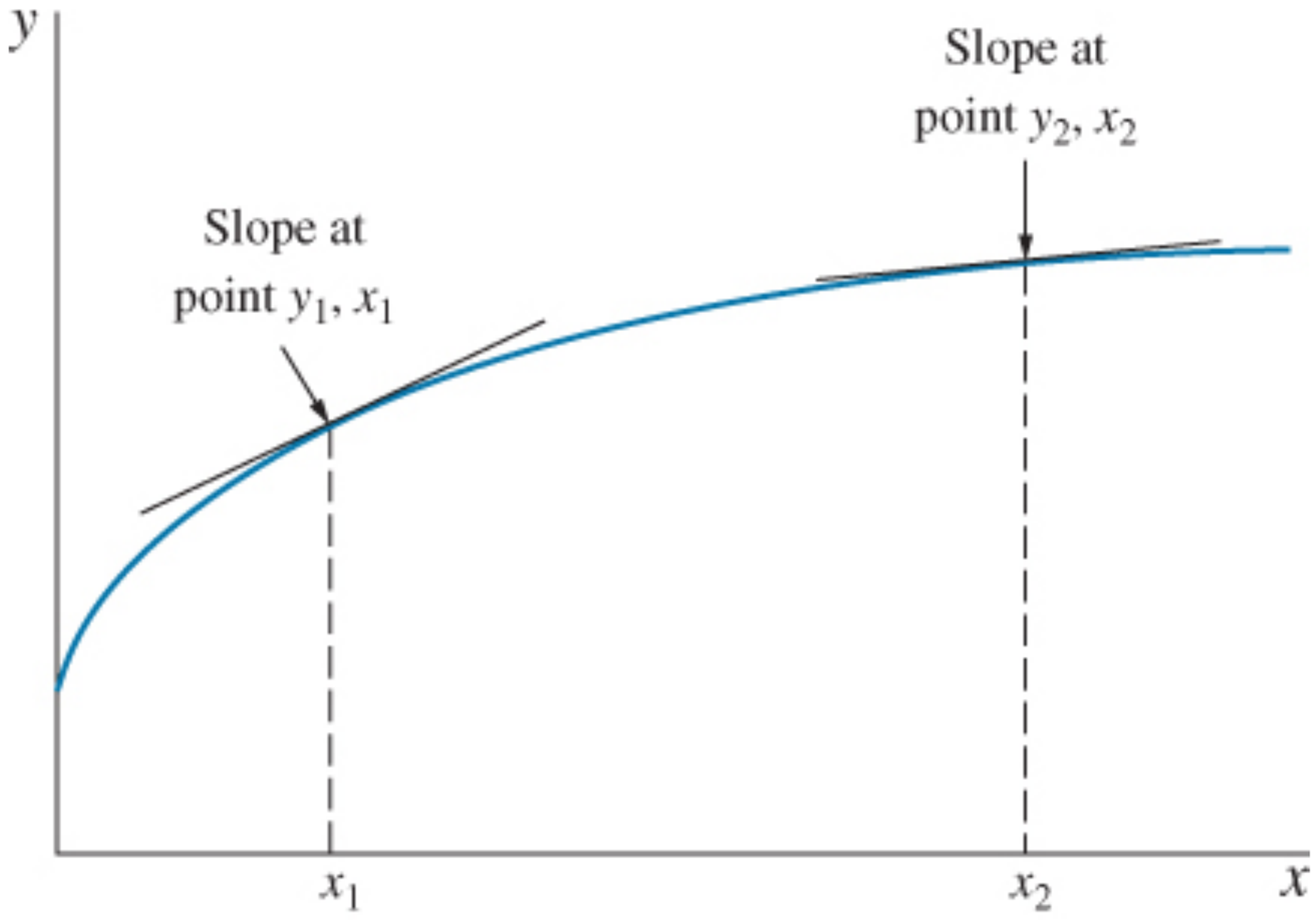
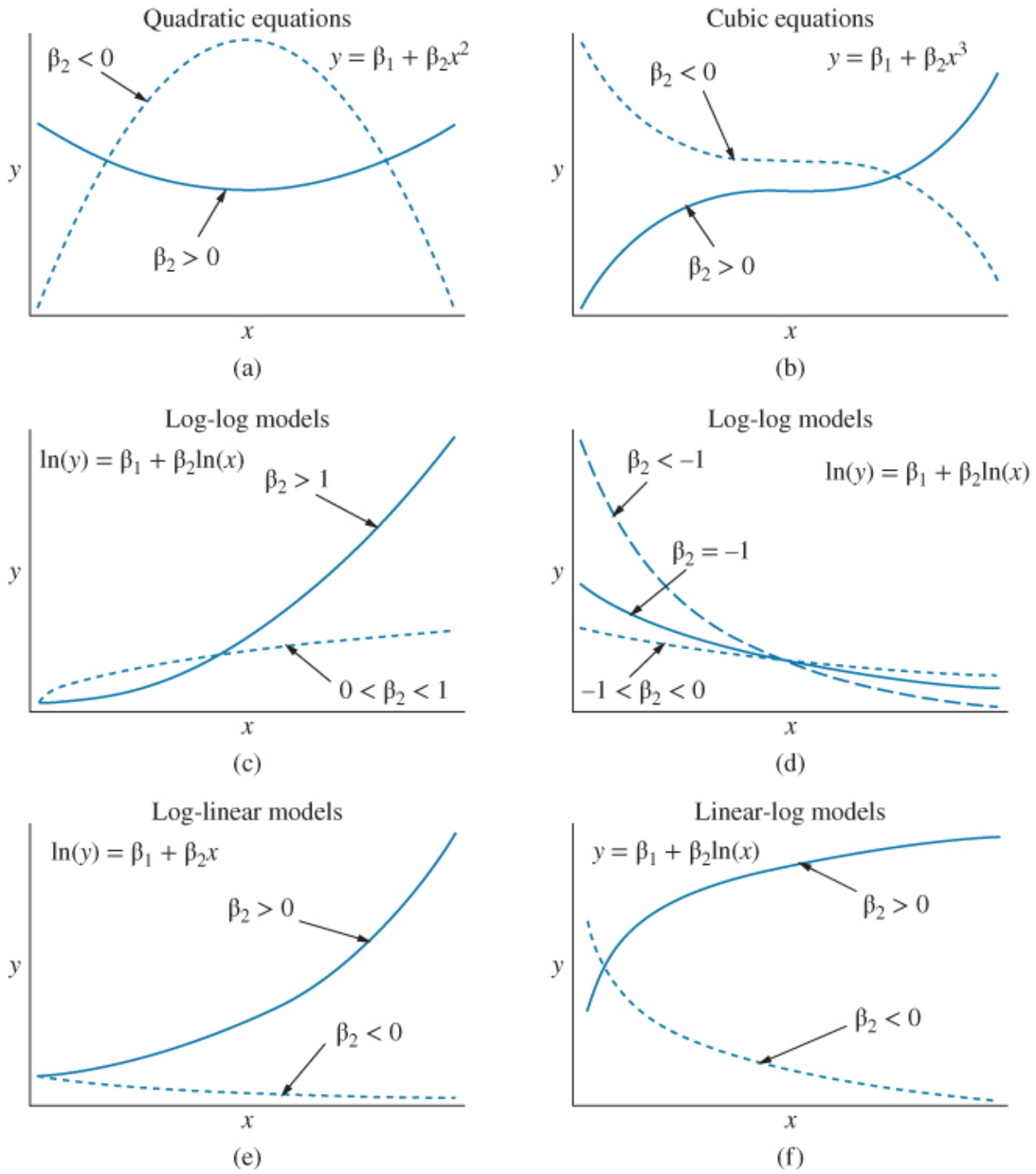


Fig 4.5 Alternative functional forms



Various models

$$\varepsilon = \frac{dy/y}{dx/x} = m\frac{x}{y}$$

$$\frac{dy/y}{dx} = \frac{m}{y}$$

$$\frac{dy}{dx/x} = mx$$

Name	Model	slope = dy/dx	Elasticity
Linear	$y = \beta_1 + \beta_2 x + e$	β_2	$\beta_2 x/y$
Quadratic	$y = \beta_1 + \beta_2 x^2 + e$	$2\beta_2 x$	$(2\beta_2 x)x/y$
Cubic	$y = \beta_1 + \beta_2 x^3 + e$	$3\beta_2 x^2 + e$	$(3\beta_2 x^2)x/y$
Log-Log	$\ln(y) = \beta_1 + \beta_2 \ln(x) + e$	$\beta_2 y/x$	β_2
Log-Linear	$\ln(y) = \beta_1 + \beta_2 x + e$	$\beta_2 y$	$\beta_2 x$
	A 1-unit increase in x leads to approximately a $100 \times \beta_2$% change in y		
Linear-Log	$y = \beta_1 + \beta_2 \ln(x) + e$	β_2/x	β_2/y
	1 % change in x leads to approximately a $\beta_2/100$ unit change in y		

Log-log: we start from $y = \exp(\beta_1 + \beta_2 \log x)$

$$\frac{dy}{dx} = \beta_2 \frac{1}{x} exp(\beta_1 + \beta_2 \log x)$$

Avoid model misspecification

When specifying a regression model, we may inadvertently choose an inadequate or incorrect functional form. How to avoid this? We check *residuals*

1. Informal check using visualization tools (subjective providing more information)
   ① residual plots (scatter plots of residuals to check if there are discernible patterns, homoskedacity)
   ② histogram, density plot, QQ plot (check normality)
2. Formal check using statistical test (adjective yet specific and narrow in conclusion).

Jarque-Bera test (H_0 : data is normally distributed, i.e,. $s = 0$, $\kappa = 3$).

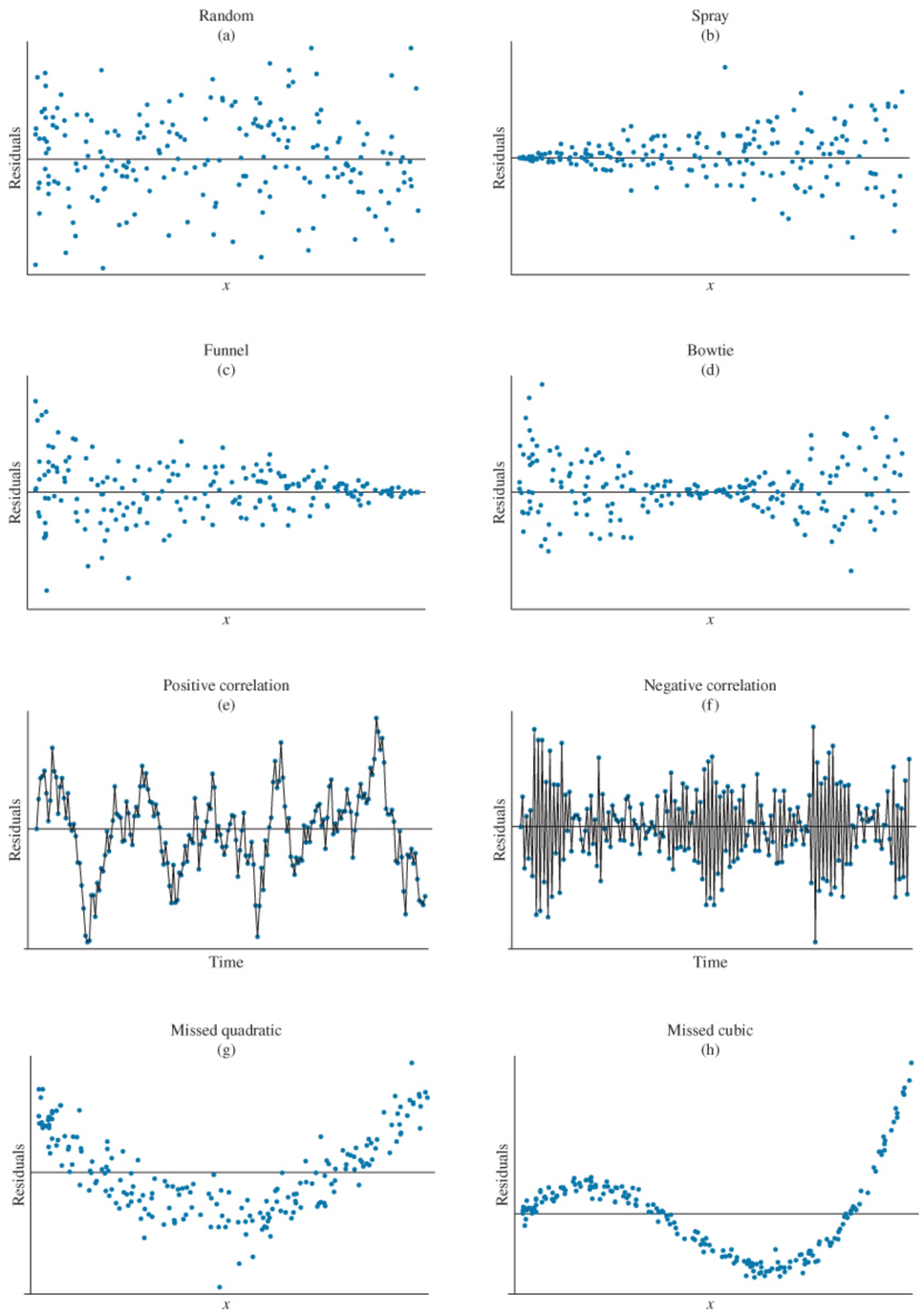
$$JB = \frac{N}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \sim \chi^2_2,$$

$$s = \frac{E[X - E[X]]^3}{\sqrt{Var(X)}^3}$$

$$\kappa = \frac{E[X - E[X]]^4}{\sqrt{Var(X)}^4}$$

where N is ample size, S is skewness, and K is kurtosis.

Examples of residuals plots



R demo: the linear-log model in food i

One can notice that the spread of the residuals seems to be higher at higher incomes, which may indicate that the homoskedasticity assumption is violated.

```
mod2 <- lm(food_exp~log(income), data=food)
ehat <- mod2$residuals
plot(food$income, ehat, xlab="income", ylab="residuals")
```

R demo: the linear-log model in food ii

Consider a simple mode: $y = 1 + x + e$, create artificial dataset, and check residuals.

```
#set.seed(12345)   #sets the seed for the random number generato
x <- runif(300, 0, 10)
e <- rnorm(300, 0, 1)
y <- 1+x+e
mod3 <- lm(y~x)
ehat <- resid(mod3)
plot(x,ehat, xlab="x", ylab="residuals")
```

R demo: the linear-log model in food iii

Consider a different model: $y = 15 - 4x^2 + e$, create artificial dataset, fit a linear regression model, and check residuals.

```
# set.seed(12345)
x <- runif(1000, -2.5, 2.5)
e <- rnorm(1000, 0, 4)
y <- 15-4*x^2+e
mod3 <- lm(y~x)
ehat <- resid(mod3)
ymi <- min(ehat)
yma <- max(ehat)
plot(x, ehat, ylim=c(ymi, yma),
     xlab="x", ylab="residuals",col="grey")
```

4.3.3 Linear-log model

- Slope or marginal effect. Note that for a linear-log model, the slope is

$$m = \frac{dy}{dx} = \frac{\beta_2}{x}.$$

- Elasticity. The elasticity is

$$\varepsilon = \frac{dy/y}{dx/x} = m\frac{x}{y} = \frac{\beta_2}{y}.$$

- Semi-elasticity. Look at the following identity:

$$\frac{dy}{100dx/x} = \frac{(dy/dx)x}{100} = \beta_2/100.$$

Ex 4.4

Let us estimate a linear-log model, $y = \beta_1 + \beta_2 ln(x) + e$, for the food dataset, draw the regression curve, and calculate the marginal effects for some given values of the dependent variable.

The estimated linear-log model using the food expenditure data is

$$\widehat{FOOD_EXP} = -97.19 + 132.17 \ln(INCOME) \quad R^2 = 0.357$$
$$\text{(se)} \qquad (84.24) \quad (28.80)^{***} \qquad (4.14)$$

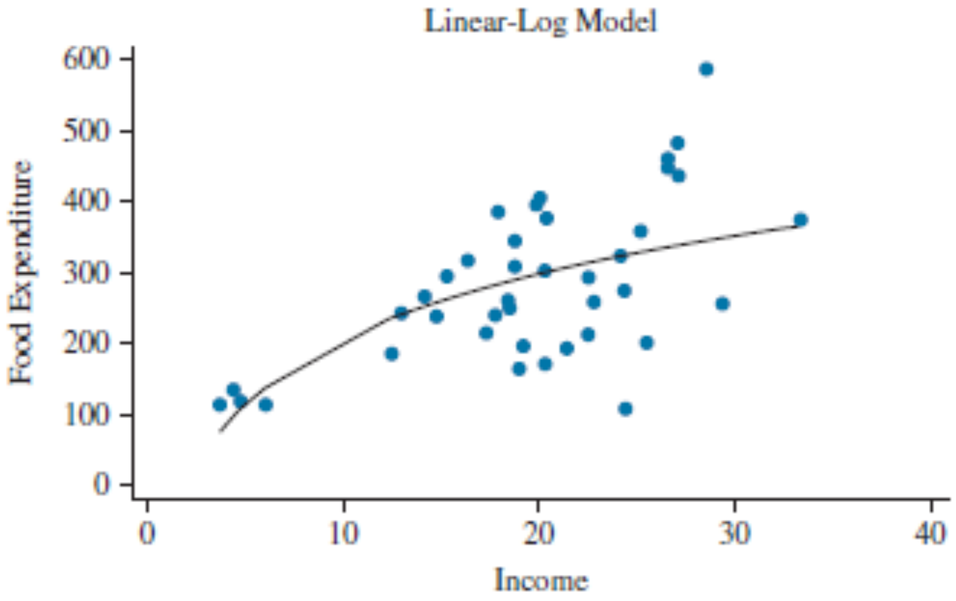


FIGURE 4.6 The fitted linear-log model.

R scripts

```
>data(food)
>mod2 <- lm(food_exp~log(income), data=food)
>b1 <- coef(mod2)[[1]]
> b2 <- coef(mod2)[[2]]
> x <- 10 #for a household earning #1000 per week
> y <- b1+b2*log(x)
> DyDx <- b2/x    #marginal effect
> DyPDx <- b2/100 #1% change in x leads to (beta2/100)-unit change in y.
> PDyPDx <- b2/y  #elasticity
> DyDx
[1] 13.21658
> DyPDx
[1] 1.321658
> PDyPDx
[1] 0.638061
```

Interpretations

We interpret the above by that a $1\ \%$ change in x leads to (approximately) a $\beta_2/100$ unit change in y.

$dy/dx = 13.217$, which indicates that an increase in income of \$100 (i.e., one unit of x) increases expenditure approximately by \$13.217;

For a $1\ \%$ increase in income (that is, an increase of \$10), expenditure increases approximately by \$1.322;

For a $1\ \%$ increase in income expenditure increases approximately by $0.638\ \%$.

Ex 4.5

Both linear and linear-log model exhibit heteroskedastic errors (unequal variances).

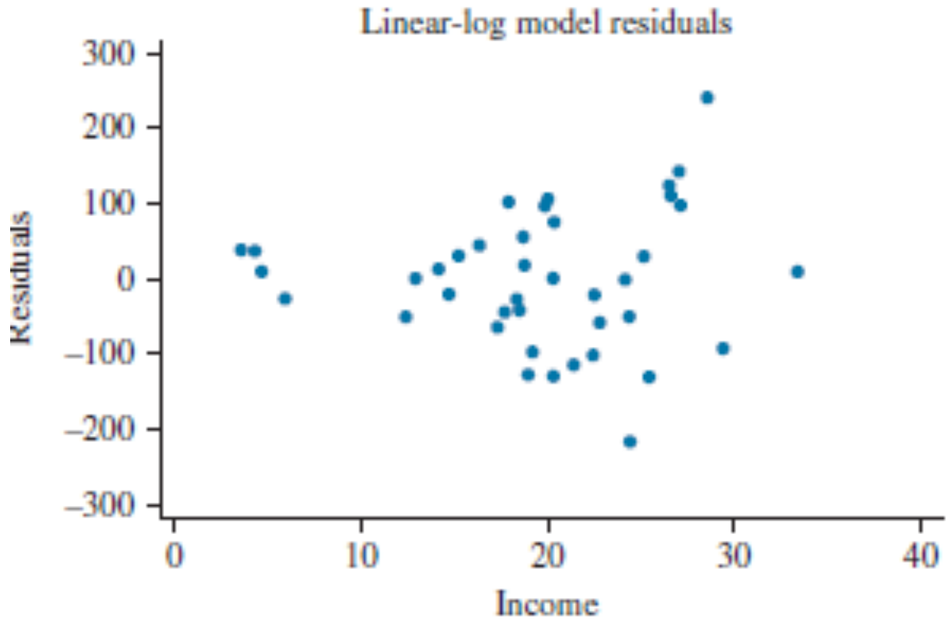
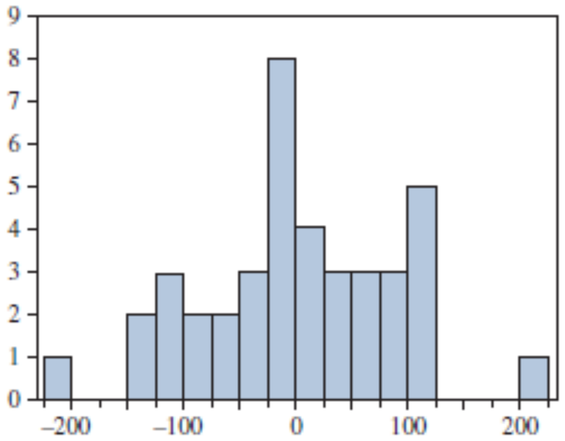


FIGURE 4.8 **Residuals from linear-log food expenditure model.**

Ex 4.6 The Jarque-Bera statistic

While the histogram may not strongly support one conclusion or another about the normality of ehat, the *Jarque-Bera test* is unambiguous.



Series: Residuals	
Sample 140	
Observations 40	
Mean	6.93e-15
Median	–6.324473
Maximum	212.0440
Minimum	–223.0255
Std. Dev.	88.36190
Skewness	–0.097319
Kurtosis	2.989034
Jarque–Bera	0.063340
Probability	0.968826

FIGURE 4.9 **EViews output: residuals histogram and summary statistics for food expenditure example.**

There is no evidence against the normality because the $p - value$ is 0.9622.

R: need `tseries` package.

R scripts

```
> library(tseries)
> ebar <- mean(ehat)
> sde <- sd(ehat)
> qqnorm(ehat)
> hist(ehat, col="grey", freq=FALSE, main="",
+      ylab="density", xlab="ehat")
> curve(dnorm(x, ebar, sde), col=2, add=TRUE,
+       ylab="density", xlab="ehat")
> jarque.bera.test(ehat) #(in package 'tseries')
Jarque Bera Test
data:  ehat
X-squared = 0.076997, df = 2, p-value = 0.9622
```

4.4 Polynomials models

Quadratic and cubic equations

The general form of a quadratic equation is

$$y = a_0 + a_1 x + a_2 x^2 \,.$$

The general form of a cubic equation is

$$y = a_0 + a_1 + a_2 x^2 + a_3 x^3 \,.$$

Ex 4.8: A growth model

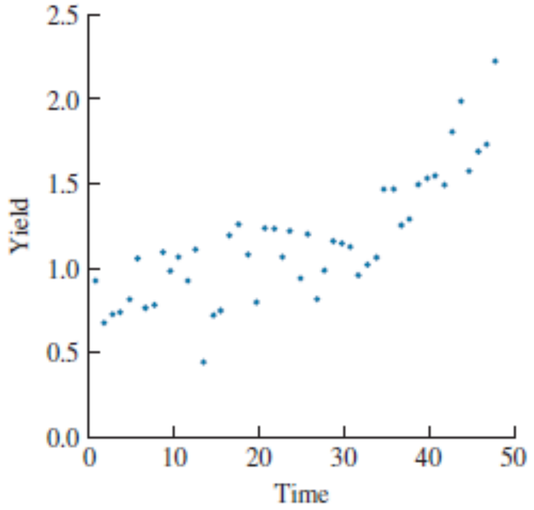


FIGURE 4.10 Scatter plot of wheat yield over time.

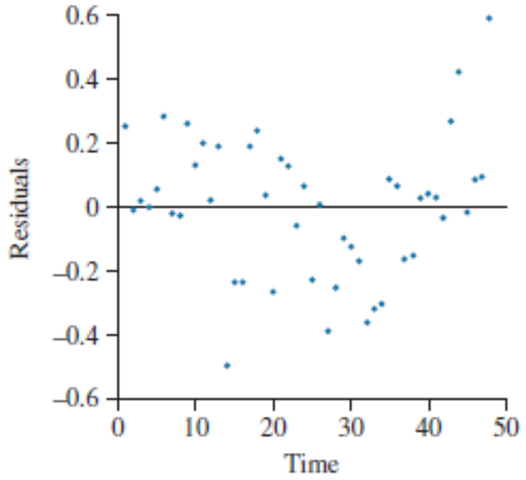


FIGURE 4.11 Residuals from a linear yield equation.

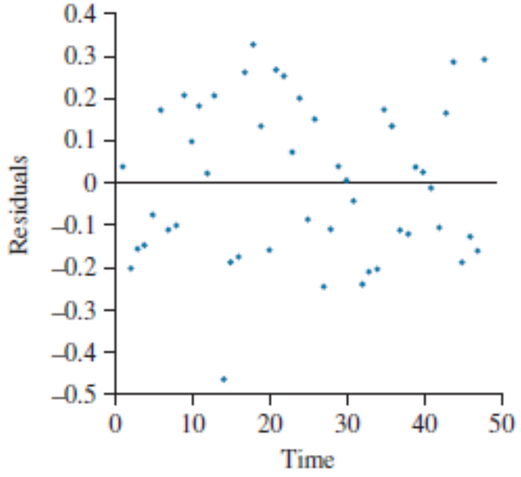


FIGURE 4.12 Residuals from a cubic yield equation.

Example: A cubic model using wa_wheat data

The ‘wa_wheat’ dataset gives annual wheat yield in tonnes per hectare in Greenough Shire in Western Australia over a period of 48 years

Since there is a pattern in the residuals. Consider a cubic model.

```
data("wa_wheat")
mod1 <- lm(greenough~time, data=wa_wheat)
ehat <- resid(mod1)
mod2 <- lm(wa_wheat$greenough~I(time^3), data=wa_wheat)
```

4.5 Log-linear models

The log-linear model

The log-linear model:

$$\ln(y) = \beta_1 + \beta_2 x\,.$$

A natural choice for prediction is

$$\hat{y}_n = \exp(b_1 + b_2 x),$$

where the subscript n is for *natural* .

A better alternative is

$$\hat{y}_c = \exp(b_1 + b_2 x + \hat{\sigma}^2/2) = \hat{y}_n \exp(\hat{\sigma}^2/2),$$

because of the property of log-normal distribution (see next page).

Appendix: Log-normal distribtuion

Suppose $Y \sim N(\mu, \sigma^2)$. Consider $W = \exp(Y)$. Then, W is said to have a log-normal distribution with mean μ and variance σ^2. We can show that

$$E(W) = \mathrm{e}^{\mu+\sigma^2/2},$$

and

$$var(W) = \mathrm{e}^{2\mu+\sigma^2}(e^{\sigma^2} - 1)\,.$$

A better prediction for the log-linear model

Then, for a log-linear model, $\ln(y) = \beta_1 + \beta_2 x + e$ with $e \sim N(0,\sigma^2)$, then

$$E(y_i) = E(\mathrm{e}^{\beta_1+\beta_2 x_i+e_i}) = E(\mathrm{e}^{\beta_1+\beta_2 x_i}\mathrm{e}^{e_i}) = \mathrm{e}^{\beta_1+\beta_2 x_i}\mathrm{e}^{\sigma^2/2} = \mathrm{e}^{\beta_1+\beta_2+\sigma^2/2}.$$

Therefore, we predict $E(y)$ with the corrected formula:

$$E(y_i) = \mathrm{e}^{b_1+b_2 x_i+\hat{\sigma}^2/2}.$$

Slope, elasticity

In a log-linear model, $\log(y) = \beta_1 + \beta_2 x + e$, focus on $y = \exp(\beta_1 + \beta_2 x)$.

The slope is

$$m = \frac{dy}{dx} = \beta_2 y\,.$$

The elasticity is

$$\varepsilon = m\frac{x}{y} = \beta_2 x\,.$$

Semi-elasticity

While x changes 1 unit, the percentage change in y is

$$\frac{100\frac{dy}{y}}{dx} = 100\frac{dy}{dx}\frac{1}{y} = 100\beta_2\,.$$

We interpret this as a 1-unit change in x leads to approximately a $100\beta_2\,\%$ change in y.

Ex 4.8: A growth model

Suppose that the yield in year t is $YIELD_t = (1 + g)YIELD_{t-1}$,

with g being a fixed growth rate in 1 year. By substituting repeatedly, we obtain

$$YIELD_t = YIELD_0(1 + g)^t .$$

Taking logarithm, we obtain

$$\ln(YILED_t) = \ln(YIELD_0) + \ln(1 + g)t = \beta_1 + \beta_2 t .$$

This is simply a log-linear model with dependent variable $\ln(YIELD_t)$ and explanatory variable t, or time.

Since $\ln(1 + g) \approx g$ while g is small, β_2 is interpreted as the growth rate.

A growth model (cont.)

Consider the log-linear model: $log(y) = \beta_1 + \beta_2 x + e$. Let us calculate the prediction: $\hat{y}_n$ and $\hat{y}_c$, marginal effect, and semi-elasticity.

Because b2 = 0.017844, we estimate that the rate of growth in wheat production has increased at an average rate of approximately 1.78 percent per year.

R scripts

```
> data(wa_wheat)
> mod4 <- lm(log(greenough)~time, data=wa_wheat)
> smod4 <- summary(mod4)
> tbl <- data.frame(xtable(smod4))
> kable(tbl, caption="Log-linear model for the *yield* equation"

|            |   Estimate| Std..Error|   t.value| Pr...t..|
|:-----------|----------:|----------:|---------:|--------:|
|(Intercept) | -0.3433665|  0.0584042| -5.879140|    4e-07|
|time        |  0.0178439|  0.0020751|  8.599107|    0e+00|
```

Example: the wage model

- The predictions and the slope are calculated for educ=12 years.
- Here are the results of these calculations: "natural" prediction $y_n = 14.796$; corrected prediction, $y_c = 16.996$; growth rate g=9.041; and marginal effect $dy/dx = 1.34$.
- The growth rate indicates that an increase in education by one unit increases hourly wage approximately by 9.041 % . (Or, an additional year of education leads to approximately 9.941 % increases in wages.)

R scripts

```
> data("cps4_small")
> xeduc <- 12
> mod5 <- lm(log(wage)~educ, data=cps4_small)
> data("cps4_small")
> xeduc <- 12
> mod5 <- lm(log(wage)~educ, data=cps4_small)
> smod5 <- summary(mod5)
> tabl <- data.frame(xtable(smod5))
> kable(tabl, caption="Log-linear 'wage' regression output")
```

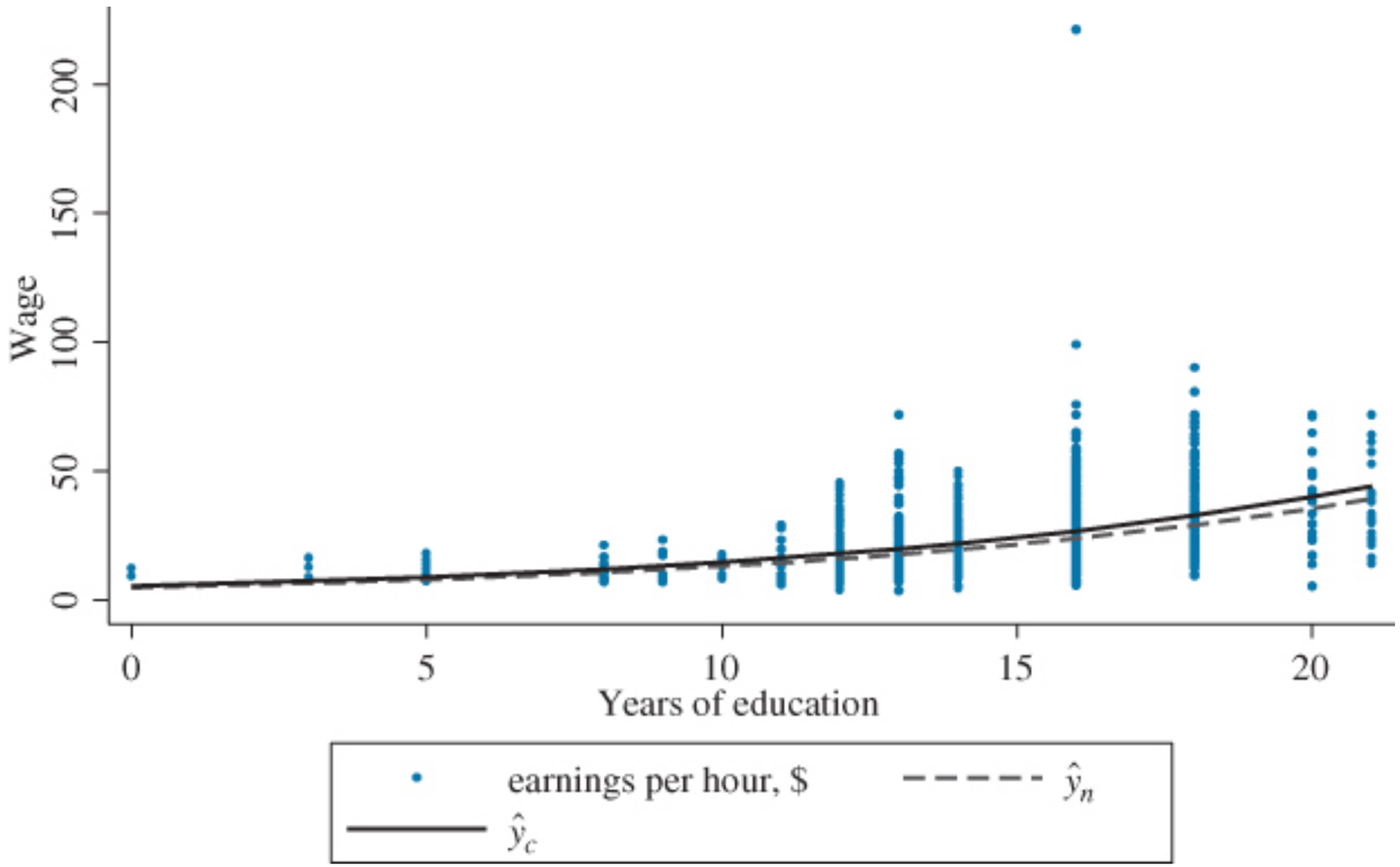
R scripts

```
|            |  Estimate| Std..Error|  t.value| Pr...t..|
|:-----------|---------:|----------:|--------:|--------:|
|(Intercept) | 1.6094445|  0.0864229| 18.62288|        0|
|educ        | 0.0904082|  0.0061456| 14.71102|        0|
```

R scripts

```
> b1 <- coef(smod5)[[1]]
> b2 <- coef(smod5)[[2]]
> sighat2 <- smod5$sigma^2
> yhatn <- exp(b1+b2*xeduc) #"natural" predictiction
> yhatc <- exp(b1+b2*xeduc+sighat2/2) #corrected prediction
> DyDx <- b2*yhatn  #marginal effect
> yhatn
[1] 14.7958
> yhatc
[1] 16.99643
> b2
[1] 0.09040825
> DyDx
[1] 1.337662
```

Fig 4.13 The natural and corrected predictors of wage



Generalized R^2

The regular R^2 cannot be used to compare two regression models having different dependent variables such as a linear-log and a log-linear models.

When such a comparison is needed, one can use the generalized R^2, which is $R_{2g} = corr(y, \hat{y})^2$

The quadratic model yields $R_{2g} = 0.188$, and the log-linear model yields $R_{2g} = 0.186$;

Since the former is higher, we conclude that the quadratic model is a better fit to the data than the log-linear one.

Note: using generalized R^2 is just one criterion for model selection. Different criteria may give different suggestions.

R scripts

```
> mod4 <- lm(wage~I(educ^2), data=cps4_small)
> yhat4 <- predict(mod4)
> mod5 <- lm(log(wage)~educ, data=cps4_small)
> smod5 <- summary(mod5)
> b1 <- coef(smod5)[[1]]
> b2 <- coef(smod5)[[2]]
> sighat2 <- smod5$sigma^2
> yhat5 <- exp(b1+b2*cps4_small$educ+sighat2/2)
> rg4 <- cor(cps4_small$wage, yhat4)^2
> rg5 <- cor(cps4_small$wage,yhat5)^2
> rg4
[1] 0.1881762
> rg5
[1] 0.1859307
```

The prediction interval in the log-linear model

A log-linear model $\log(y) = \beta_1 + \beta_2 x + e$ equals $w = \beta_1 + \beta_2 + e$, where $w = \log(y)$ is the log-transformation of y.

To forecast y, we first forecast w and obtain its prediction interval

$$(b_1 + b_2 x) \pm t_c se(f),$$

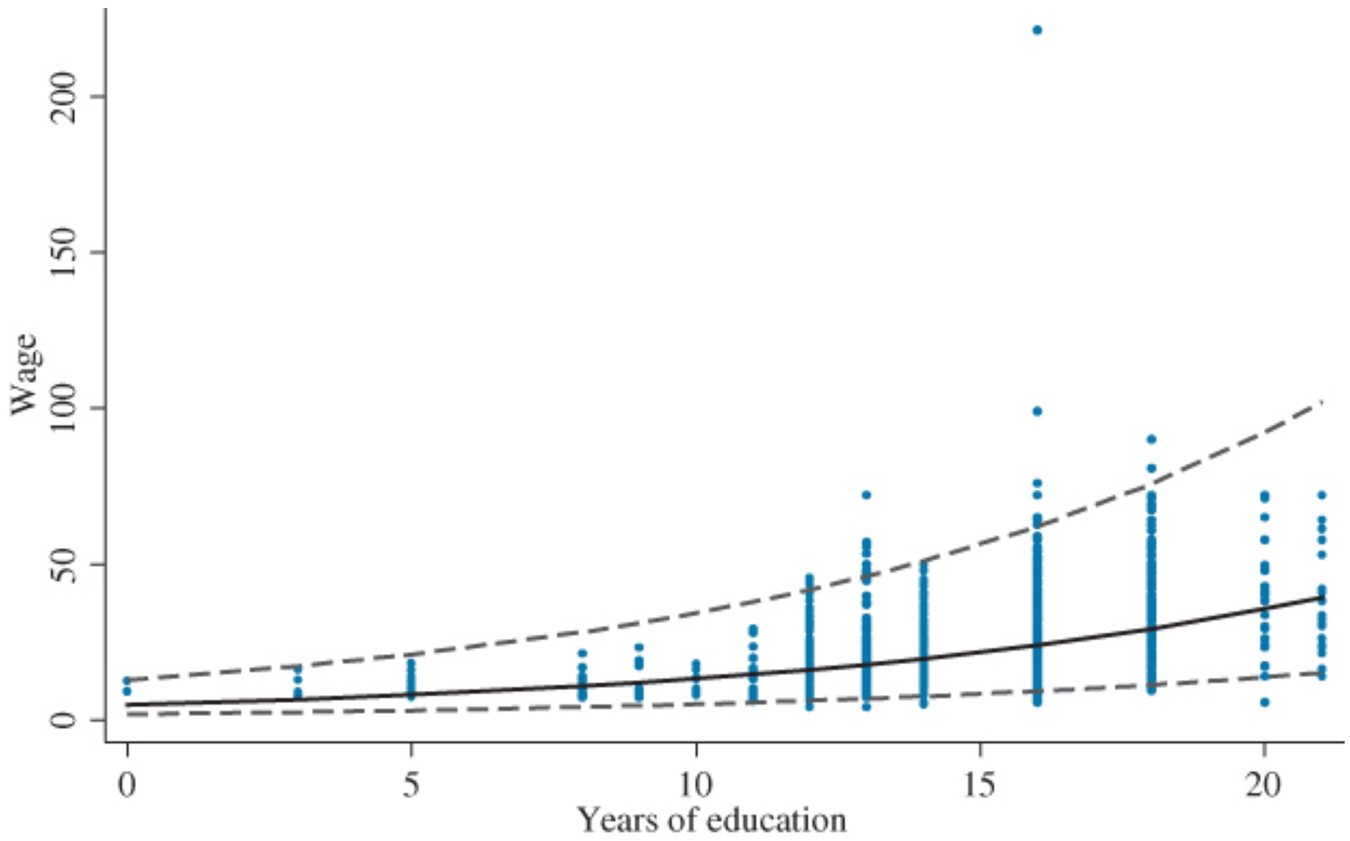
where $se(f)$ can be calculated as the usual simple linear regression.

Then we took exponential for every part to obtain the prediction interval for a forecast in y,

$$exp((b_1 + b_2 x) \pm t_c se(f)) .$$

Fig 14. The 95% prediction intervals in the log-linear model for the wage

The prediction interval for a forecast of *y* while educ = 12 is (5.26,41.62). We can also draw a 95% prediction band for the log-linear wage model.



R scripts i

Find prediction interval estimate for educ = 12.

```
> alpha <- 0.05 > xeduc <- 12
> xedbar <- mean(cps4_small$educ)
> mod5 <- lm(log(wage)~educ, data=cps4_small)
> b1 <- coef(mod5)[[1]]
> b2 <- coef(mod5)[[2]]
> df5 <- mod5$df.residual
> N <- nobs(mod5)
> tcr <- qt(1-alpha/2, df=df5)
> smod5 <- summary(mod5)
> varb2 <- vcov(mod5)[2,2]
> sighat2 <- smod5$sigma^2
> varf <- sighat2+sighat2/N+(xeduc-xedbar)^2*varb2
\framebreak
```

R scripts ii

```
> sef <- sqrt(varf)
> lnyhat <- b1+b2*xeduc
> lowb <- exp(lnyhat-tcr*sef)
> upb <- exp(lnyhat+tcr*sef)
> lowb
[1] 5.260398
> upb
[1] 41.61581
```

4.6 Log-log models

Slope and elasticity

The log-log-function, $\ln(y) = \beta_1 + \beta_2 \ln(x)$, is widely used to describe demand equations and production functions.

We focus on $y = exp(\beta_1 + \beta_2 \ln(x))$.

The first-order differentiation (slope) is

$$m = \frac{dy}{dx} = \frac{\beta_2 \exp(\beta_1 + \beta_2 \ln(x))}{x} = \frac{\beta_2 y}{x}\,.$$

The elasticity is β_2,

$$\varepsilon = m\frac{x}{y} = \frac{\beta_2 y}{x}\frac{x}{y} = \beta_2\,.$$

The second-order differentiation

The second-order differentiation is

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{\beta_2^2 \exp(\beta_1 + \beta_2 \ln(x)) - \beta_2 \exp(\beta_1 + \beta_2 \ln(x))}{x^2} \\ &= \frac{\beta_2 \exp(\beta_1 + \beta_2 \ln(x))(\beta_2 - 1)}{x^2}.\end{aligned}$$

1. If $\beta_2 > 0$, then y is an increasing function of x at an an increasing rate for $\beta_2 > 1$ but at a decreasing rate for $0 < \beta_2 < 1$.
2. If $\beta_2 < 0$, there is an inverse relationship between y and x.

Prediction in the log-log model

Similar to the log-linear model, a natural choice for prediction (or the fitted value) is

$$\hat{y}_n = \exp(b_1 + b_2 \log(x)) \,.$$

A better alternative is

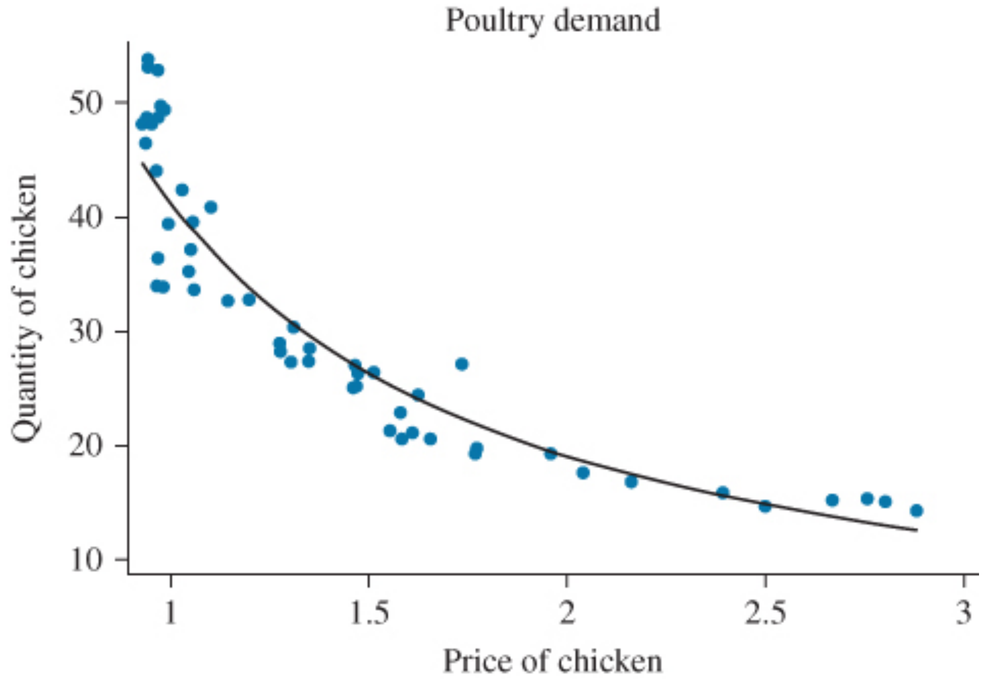
$$\hat{y}_c = \exp(b_1 + b_2 \log(x) + \hat{\sigma}^2/2) \,.$$

Ex 4.13:

The log-log functional form is frequently used for demand equations. Consider, for example, the demand for edible chicken, which the U.S. Department of Agriculture calls "broilers." The data for this exercise are in the data file *newbroiler*, which is adapted from the data provided by Epple and McCallum (2006).[2] The scatter plot of Q = per capita consumption of chicken, in pounds, versus P = real price of chicken is shown in Figure 4.15 for 52 annual observations, 1950–2001. It shows the characteristic hyperbolic shape that was displayed in Figure 4.5(d).

The estimated log-log model is

$$\widehat{\ln(Q)} = 3.717 - 1.121 \times \ln(P) \qquad R_g^2 = 0.8817 \qquad (4.15)$$
$$\text{(se)} \quad (0.022) \quad (0.049)$$



Example: calculating log-log demand for chicken

The coefficient on p indicates that an increase in price by $1\ \%$ changes the quantity demanded by $-1.121\ \%$.

The generalized $R - squared$ is 0.88.

We can also draw the fitted values of the log-log equation by using the corrected formula of the fitted value.

```
> data("newbroiler", package="PoEdata")
> mod6 <- lm(log(q)~log(p), data=newbroiler)
```